



Download Swarachakra

The Swarachakra keyboard is available in 12 languages, and is designed by the Industrial Design Center at IIT-B <https://digit.in/sep19-45>



Designing Swarachakra

Anirudha Joshi who teaches interaction design at IIT-B explains the design process of Swarachakra <https://digit.in/sep19-46>

is that almost anybody who used an Indic language keypad for the first time, did not have any difficulty finding out how to type conjuncts.” This special key allows users to modify the direction in which the consonants get conjuncted, or prevent two consonants from conjuncting. Another useful feature on Swalekh Flip is a joystick, to navigate through the text. This is because the cursor also carries the legacy and baggage of being designed for English. Imagine a cursor moving through **स्त्र्याजीव** (stryajeev) or **हृदरोग** (hridrog) and backspacing through these words. It does not work the way you are imagining it. The cursor behaves in different and erratic ways



The Swarachakra layout for Oriya

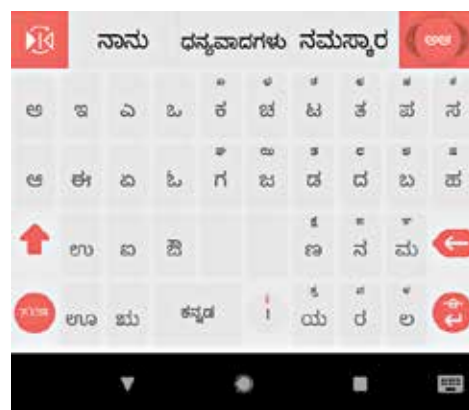
depending on the application for even simple functions such as backspace or delete. Pani explains “In conjuncts, when matras join a consonant, it becomes difficult to differentiate what is where. Therefore, it becomes very difficult for a user to place the cursor select a piece of text, or delete, edit and do minor corrections that they want on the text. That is the reason why we introduced the joystick. The user can have finer control over the cursor and editing and all that”. They keyboard also allows users to set the font size, and toggle the visibility of the borders around the keys.

Swarachakra is another Indic keyboard that has a rather unique input mechanism. Every time a consonant is typed, the possible matras are shown in a circular disc around the letter,

allowing users to quickly add the matras. This means that keying in any part of the barakhadi is a single action, instead of the two it typically takes on most other keyboards. The user touches a consonant, and then navigates to the appropriate matra. Right at the top is the halant, if the user navigates to that, then all the possible conjuncts for the particular character shows up on the keypad. The vowels appear in their own circle. The app has been developed by IIT-B.

Swarachakra uses an innovative approach of sorting the keys by varga, but having only five keys in each row. This understandably means that there are 8 rows, but the user can adjust the heights of each of the rows, and choose how much of the screen the keyboard should occupy. The width of each of the keys means that instead of the usual ten keys per row, there is very little room for errors, which improves the typing experience.

Indus OS uses a proprietary keyboard called Indus Swipe, parts of which are patented. There are 12 languages supported, Assamese, Bengali, Gujarati, Hindi, Kannada, Malayalam, Marathi, Odia, Punjabi, Telugu, Tamil and Urdu. Like the rest of the OS, the keyboard was designed to be used by people who may not be familiar with the technologies, and may even be using a smartphone keyboard for the first time, and designed specifically for users who did not know how to type in English. The design approach did not follow the lines of a QWERTY keyboard for English, something that originated from typewriter keyboards. The keyboards available when Indus Swipe was being developed, in 2014 could not fit all the characters of an Indic language onto a single screen. In the Indus Swipe keyboard as well, the arrangement of the keys is based on the varnamala taught in school, and is based on how most people learn the vernacular languages. The consonants are placed in two blocks of five characters, as they are taught in schools.



The Swalekh Flip keyboard

The last character in these sequences are marked with a slightly darker key, making it easy for users to intuitively find the characters that they are going for. There are ten characters in each row. The vowels are strung up at the bottom of the keyboard. Another aspect of easing the layout, is accurate predictions of what the user wants to type. The keyboards from Indic languages stand to benefit from being dynamic, where the keys available to the user change based on the text that has already been entered. We will come back to that.

OPTIMISATION

Gboard allows users to download the additional languages according to their needs. There is also granular control over which layouts, scripts and input methods they want to support for particular languages. The size of the application varies with the device and the selected modules. Swarachakra has individual apps for each of the languages that it supports. Swalekh Flip has a size of 9 MB, for all the 22 supported languages, which is phenomenal considering its capabilities. The keyboard was actually Smart-Pad when it was first released in 2011. The memory constraints were harsh at that time, with even a 2 MB application appearing large. Pani explains, “our objective in that point in time was to make Swalekh such that every Indian should be able to use their keypad, without thinking that it is going to load their machines very much, and